

ASSESSMENT TOOL – 4

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COURSE NAME: Cloud Computing and Big Data Analysis

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.

ANSWER:

Feature	Type-1 Hypervisor (Bare Metal)	Type-2 Hypervisor (Hosted)
Installation	Runs directly on hardware	Runs on top of a host operating system
Performance	High performance, low overhead	Lower performance due to OS layer
Security	More secure, smaller attack surface	Less secure because host OS can be attacked
Reliability	Highly reliable	Depends on host OS stability
Manageability	Centralized management, live migration, clustering	Limited management features
Examples	VMware ESXi, Microsoft Hyper-V, Xen	Oracle VirtualBox, VMware Workstation

Analysis

- **Performance:** Type-1 provides near-native performance and supports large workloads efficiently.
- **Security:** Since there is no host operating system, attackers have fewer opportunities to compromise the system.

- **Manageability:** Supports advanced cloud features such as live migration, high availability (HA), fault tolerance, and automated resource allocation.

Justification

Type-1 Hypervisor is the better choice for mission-critical cloud applications because it offers:

- Better performance
- Stronger security
- Higher reliability
- Easier centralized management
- Better scalability for enterprise cloud environments

2.A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

ANSWER:

A cloud provider hosts:

- Finance applications
- Healthcare applications
- Student portals

All run on the same physical server but inside separate Virtual Machines (VMs).

How Virtualization Supports Isolation

- Each VM has its own:
 - Operating System
 - CPU allocation
 - Memory space
 - Virtual storage
 - Virtual network
- The hypervisor ensures one VM cannot directly access another VM's resources.

- Network virtualization separates traffic using VLANs or virtual switches.
- Storage virtualization allocates dedicated virtual disks to each workload.

Benefits

- Prevents data leakage.
- Improves security.
- Allows independent resource allocation.
- Simplifies compliance with regulations (HIPAA, PCI-DSS).

Two Major Attack Surfaces

1. Hypervisor Attacks

- Vulnerabilities in the hypervisor may allow attackers to escape from one VM into another.
- Known as VM Escape.

2. Virtual Network Attacks

- Attackers may intercept traffic between VMs.
- Examples:
 - ARP spoofing
 - Virtual switch attacks
 - Packet sniffing

Conclusion

Virtualization provides strong isolation by separating compute, storage, and networking resources while protecting workloads through hypervisor-enforced boundaries.

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

ANSWER:

Resource	Value
CPU Utilization	92%

Resource	Value
Memory Utilization	88%
Storage I/O Latency	35 ms
Average VM Boot Time	170 s

Analysis:

CPU = 92%

- CPU is heavily utilized.
- VMs may experience slow processing and scheduling delays.

Memory = 88%

- Memory is nearly full.
- Hypervisor may start memory swapping or ballooning.

Storage Latency = 35 ms

- Very high latency.
- Indicates storage subsystem bottleneck.
- VM disk operations become slow.

VM Boot Time = 170 s

- Normal VM boot is around 30–60 seconds.
- Slow boot indicates storage and CPU bottlenecks.

Likely Bottlenecks

- CPU resource contention
- Memory pressure
- Slow disk I/O
- Overloaded storage array

Corrective Actions

- Increase CPU cores or migrate VMs to another host.
- Add more RAM.

- Upgrade storage to SSD/NVMe.
- Balance workloads across multiple hosts.
- Enable resource monitoring and autoscaling.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

ANSWER:

Feature	Traditional Datacenter	Software Defined Datacenter
Compute	Physical servers	Virtual machines/containers
Storage	Physical disks	Software-defined storage
Network	Hardware switches	Software-defined networking (SDN)
Provisioning	Manual	Automated
Scalability	Limited	Highly scalable
Resource Allocation	Fixed	Dynamic

Compute Virtualization

- Creates virtual machines.
- Resources are allocated dynamically.
- Supports live migration and high availability.

Storage Virtualization

- Combines multiple storage devices into one logical pool.
- Enables thin provisioning and snapshots.
- Improves storage utilization.

Network Virtualization

- Creates virtual networks independent of physical hardware.
- Supports VLANs, VXLANs, and SDN.
- Enables rapid network configuration.

How SDDC Improves Agility

- Faster deployment of applications.
- Automatic resource provisioning.
- Centralized management.
- Better scalability.
- Reduced operational costs.
- Improved disaster recovery and business continuity.

Conclusion

SDDC provides greater agility than traditional datacenters by virtualizing compute, storage, and networking, allowing automated, flexible, and efficient cloud operations.

5.A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

ANSWER:

Problem

A multi-cloud project uses different cloud providers.

Each provider has:

- Different user identities
- Different authentication systems
- Different access control policies

This causes:

- Login failures
- Permission mismatches
- Poor user experience
- Security risks

Proposed Secure Identity and Privacy Design

Identity Federation

- Establish trust between cloud providers.
- Users authenticate once and access multiple clouds (Single Sign-On).

Federated Identity Management

- A central Identity Provider (IdP) authenticates users.
- Cloud providers rely on the IdP for identity verification.

Role-Based Access Control (RBAC)

- Assign permissions based on user roles (Admin, Doctor, Finance Staff, Student).

Multi-Factor Authentication (MFA)

- Require password plus OTP or biometric verification.

Privacy Protection

- Encrypt identity information during transmission.
- Share only necessary user attributes (minimum disclosure).
- Maintain audit logs for all authentication events.

Benefits

- Secure Single Sign-On (SSO)
- Consistent identity across cloud providers
- Reduced password management
- Improved privacy protection
- Better compliance with security regulations

Conclusion

Using identity federation, RBAC, MFA, and privacy-preserving authentication enables secure and seamless access across multiple cloud providers while protecting user identities and sensitive data.